

Calculus III – Five Mock Finals: Worked Solutions

Mock Final 1

Q1(a). $f(x, y) = x^2y + e^{xy}$, so

$$\nabla f = (2xy + ye^{xy}, x^2 + xe^{xy}).$$

At $(1, 0)$, $\nabla f = (0, 2)$. The direction from $(1, 0)$ to $(4, 4)$ is $\langle 3, 4 \rangle$, hence

$$\mathbf{u} = \left\langle \frac{3}{5}, \frac{4}{5} \right\rangle, \quad D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} = \boxed{\frac{8}{5}}.$$

Q1(b). Rewrite

$$g = x^2 + y^2 - 2x - 4y = (x - 1)^2 + (y - 2)^2 - 5.$$

The interior critical point is $(1, 2)$, which lies inside $x^2 + y^2 \leq 9$, giving $g = -5$. On $x^2 + y^2 = 9$, the farthest point from $(1, 2)$ is opposite the vector $\langle 1, 2 \rangle$:

$$\left(-\frac{3}{\sqrt{5}}, -\frac{6}{\sqrt{5}} \right).$$

There the distance from $(1, 2)$ is $3 + \sqrt{5}$, so

$$g_{\max} = (3 + \sqrt{5})^2 - 5 = \boxed{9 + 6\sqrt{5}}.$$

Thus

$$\boxed{g_{\min} = -5 \text{ at } (1, 2)}, \quad \boxed{g_{\max} = 9 + 6\sqrt{5} \text{ at } \left(-\frac{3}{\sqrt{5}}, -\frac{6}{\sqrt{5}} \right)}.$$

Q2(a). Use polar coordinates: $x^2 + y^2 = r^2$, $dA = r dr d\theta$, with $1 \leq r \leq 2$, $0 \leq \theta \leq \pi$:

$$\iint_D r^2 e^{-r^2} dA = \pi \int_1^2 r^3 e^{-r^2} dr = \boxed{\frac{\pi}{2} (2e^{-1} - 5e^{-4})}.$$

Q2(b). In spherical coordinates, $x^2 + y^2 + z^2 = \rho^2$ and $dV = \rho^2 \sin \phi d\rho d\phi d\theta$. For the upper half-ball:

$$0 \leq \rho \leq 2, \quad 0 \leq \phi \leq \frac{\pi}{2}, \quad 0 \leq \theta \leq 2\pi.$$

Hence

$$\iiint_E \rho^2 dV = \int_0^{2\pi} \int_0^{\pi/2} \int_0^2 \rho^4 \sin \phi d\rho d\phi d\theta = \boxed{\frac{64\pi}{5}}.$$

Q3(a). For $\mathbf{r}(t) = (2 \cos t, 2 \sin t, t)$,

$$|\mathbf{r}'(t)| = \sqrt{5}, \quad x^2 + y^2 + z = 4 + t.$$

Thus

$$\int_C (x^2 + y^2 + z) ds = \sqrt{5} \int_0^\pi (4 + t) dt = \boxed{\sqrt{5} \left(4\pi + \frac{\pi^2}{2} \right)}.$$

Q3(b). $P = x - y$, $Q = x + y$, so

$$Q_x - P_y = 1 - (-1) = 2.$$

The region is $0 \leq x \leq 2$, $x^2 \leq y \leq 2x$. Therefore

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^2 \int_{x^2}^{2x} 2 dy dx = \boxed{\frac{8}{3}}.$$

Q4. The cone is $z = g(x, y) = \sqrt{x^2 + y^2} = r$. For downward orientation,

$$\mathbf{n} \, dS = \langle g_x, g_y, -1 \rangle \, dA = \langle \cos \theta, \sin \theta, -1 \rangle r \, dr \, d\theta.$$

With $\mathbf{F} = \langle r \cos \theta, r \sin \theta, r^2 \rangle$,

$$\mathbf{F} \cdot \mathbf{n} \, dS = (r^2 - r^3) \, dr \, d\theta.$$

Thus

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = 2\pi \int_1^3 (r^2 - r^3) \, dr = \boxed{-\frac{68\pi}{3}}.$$

The negative sign means the net flow is opposite the chosen downward normal, i.e. predominantly upward through the cone.

Mock Final 2

Q1(a). For $f = xe^{yz} + y^2z$,

$$\nabla f = \langle e^{yz}, xze^{yz} + 2yz, xye^{yz} + y^2 \rangle.$$

At $(1, 0, 2)$ this is $\langle 1, 2, 0 \rangle$. The unit direction of $\langle 2, -1, 2 \rangle$ is $\frac{1}{3}\langle 2, -1, 2 \rangle$, so

$$D_{\mathbf{u}}f = \frac{1}{3}(2 - 2) = \boxed{0}.$$

Q1(b). Max/min of $h = xy$ subject to $x^2 + 4y^2 = 4$. Lagrange:

$$\langle y, x \rangle = \lambda \langle 2x, 8y \rangle.$$

This gives $\lambda = \pm \frac{1}{4}$ and the candidates

$$(\pm\sqrt{2}, \pm 1/\sqrt{2}), \quad (\pm\sqrt{2}, \mp 1/\sqrt{2}).$$

Therefore

$$\boxed{h_{\max} = 1}, \quad \boxed{h_{\min} = -1}.$$

Q2(a). The region satisfies $r \leq 2$ and $r \sin \theta \geq 1$, so

$$\frac{\pi}{6} \leq \theta \leq \frac{5\pi}{6}, \quad \csc \theta \leq r \leq 2.$$

Since the integrand is r and $dA = r dr d\theta$,

$$I = \int_{\pi/6}^{5\pi/6} \int_{\csc \theta}^2 r^2 dr d\theta = \frac{1}{3} \int_{\pi/6}^{5\pi/6} (8 - \csc^3 \theta) d\theta.$$

Using

$$\int_{\pi/6}^{5\pi/6} \csc^3 \theta d\theta = 2\sqrt{3} + \ln(2 + \sqrt{3}),$$

we obtain

$$\boxed{I = \frac{16\pi}{9} - \frac{2\sqrt{3}}{3} - \frac{1}{3} \ln(2 + \sqrt{3})}.$$

Q2(b). Above $z = \sqrt{x^2 + y^2}$ means $0 \leq \phi \leq \pi/4$. Thus

$$V = \int_0^{2\pi} \int_0^{\pi/4} \int_0^3 \rho^2 \sin \phi d\rho d\phi d\theta = \boxed{9\pi(2 - \sqrt{2})}.$$

Q3(a). Here

$$\mathbf{r}'(t) = \langle 1, 2t, 3t^2 \rangle, \quad ds = \sqrt{1 + 4t^2 + 9t^4} dt, \quad x = t.$$

Hence

$$I = \int_0^1 t \sqrt{1 + 4t^2 + 9t^4} dt.$$

Let $u = t^2$, $du = 2t dt$:

$$I = \frac{1}{2} \int_0^1 \sqrt{9u^2 + 4u + 1} du.$$

An exact antiderivative gives

$$\boxed{I = -\frac{1}{18} + \frac{11\sqrt{14}}{36} + \frac{5}{108} \left[\operatorname{arsinh}\left(\frac{11\sqrt{5}}{5}\right) - \operatorname{arsinh}\left(\frac{2\sqrt{5}}{5}\right) \right]}.$$

(This is deliberately the least pleasant line-integral computation in the set.)

Q3(b). $P = x^2 - y$, $Q = x + y^2$. Then

$$Q_x - P_y = 1 - (-1) = 2.$$

The rectangle has area 6, so

$$\oint_C P dx + Q dy = 2(6) = \boxed{12}.$$

Q4. $z = g(x, y) = 4 - x^2 - y^2$ and the upward area vector is

$$\langle -g_x, -g_y, 1 \rangle dA = \langle 2x, 2y, 1 \rangle dA.$$

Thus

$$\mathbf{F} \cdot \mathbf{n} dS = (2x^3 + 2y^3 + 4 - x^2 - y^2) dA.$$

Over the disk $x^2 + y^2 \leq 4$, the odd terms vanish by symmetry:

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \int_0^{2\pi} \int_0^2 (4 - r^2) r dr d\theta = \boxed{8\pi}.$$

Positive flux means net flow crosses in the chosen upward direction.

Mock Final 3

Q1(a). For $f = xe^{-y} + ye^{-x}$,

$$\nabla f = \langle e^{-y} - ye^{-x}, -xe^{-y} + e^{-x} \rangle.$$

At $(0, 0)$, $\nabla f = \langle 1, 1 \rangle$. Therefore

$$\boxed{\text{maximum rate} = \sqrt{2}}, \quad \boxed{\text{direction} = \frac{1}{\sqrt{2}}\langle 1, 1 \rangle}.$$

Q1(b). For $g = x + 2y$ on the upper half-disk $x^2 + y^2 \leq 4$, $y \geq 0$, the maximum occurs on the circular arc in the direction $\langle 1, 2 \rangle$:

$$\left(\frac{2}{\sqrt{5}}, \frac{4}{\sqrt{5}} \right), \quad g = 2\sqrt{5}.$$

On the diameter $y = 0$, $g = x$, whose minimum is -2 at $(-2, 0)$. Hence

$$\boxed{g_{\max} = 2\sqrt{5}}, \quad \boxed{g_{\min} = -2}.$$

Q2(a). Quarter disk: $0 \leq \theta \leq \pi/2$, $0 \leq r \leq \sqrt{\pi}$:

$$I = \int_0^{\pi/2} \int_0^{\sqrt{\pi}} \cos(r^2) r \, dr \, d\theta = \frac{\pi}{4} [\sin(r^2)]_0^{\sqrt{\pi}} = \boxed{0}.$$

Q2(b). By symmetry on the ball,

$$\iiint_E z^2 \, dV = \frac{1}{3} \iiint_E (x^2 + y^2 + z^2) \, dV.$$

Therefore

$$\iiint_E z^2 \, dV = \frac{1}{3} \cdot 4\pi \int_0^a \rho^4 \, d\rho = \boxed{\frac{4\pi a^5}{15}}.$$

Q3(a). Parametrize the quarter-circle by

$$\mathbf{r}(t) = \langle 2 \cos t, 2 \sin t \rangle, \quad 0 \leq t \leq \pi/2,$$

so $ds = 2dt$. Then

$$\int_C (x + y) \, ds = 4 \int_0^{\pi/2} (\cos t + \sin t) \, dt = \boxed{8}.$$

Q3(b). $P = x^3 - y$, $Q = x + y^3$, so $Q_x - P_y = 2$. Since the disk has area 4π ,

$$\oint_C P \, dx + Q \, dy = 2(4\pi) = \boxed{8\pi}.$$

Q4. Close the hemisphere with the disk $z = 0$. For

$$\mathbf{F} = \langle xz, yz, z \rangle, \quad \nabla \cdot \mathbf{F} = z + z + 1 = 2z + 1.$$

The disk contributes zero flux because $z = 0$ there. Hence the hemisphere flux is

$$\iiint_E (2z + 1) \, dV.$$

The half-ball volume is $16\pi/3$, and

$$\iiint_E z \, dV = 4\pi.$$

Thus

$$\boxed{\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = 2(4\pi) + \frac{16\pi}{3} = \frac{40\pi}{3}}.$$

Mock Final 4**Q1(a).** For $f = \ln(x^2 + y^2)$,

$$\nabla f = \left\langle \frac{2x}{x^2 + y^2}, \frac{2y}{x^2 + y^2} \right\rangle.$$

At $(1, \sqrt{3})$,

$$\nabla f = \left\langle \frac{1}{2}, \frac{\sqrt{3}}{2} \right\rangle.$$

The unit vector making angle $\pi/3$ with the positive x -axis is the same vector, so

$$D_{\mathbf{u}}f = \boxed{1}.$$

Q1(b). On $x^2 + y^2 \leq 4$,

$$g = x^2 - y^2.$$

The largest value occurs when $y = 0$, $|x| = 2$, and the smallest when $x = 0$, $|y| = 2$:

$$\boxed{g_{\max} = 4 \text{ at } (\pm 2, 0)}, \quad \boxed{g_{\min} = -4 \text{ at } (0, \pm 2)}.$$

Q2(a). In polar coordinates, $1 \leq r \leq 3$, $0 \leq \theta \leq \pi/4$ and

$$xy = r^2 \cos \theta \sin \theta.$$

Hence

$$I = \int_0^{\pi/4} \int_1^3 r^3 \cos \theta \sin \theta \, dr \, d\theta = 20 \cdot \frac{1}{4} = \boxed{5}.$$

Q2(b). For $z = r$, $\phi = \pi/4$; for $z = \sqrt{3}r$, $\phi = \pi/6$. Thus

$$0 \leq \rho \leq 4, \quad \frac{\pi}{6} \leq \phi \leq \frac{\pi}{4}, \quad 0 \leq \theta \leq 2\pi.$$

Therefore

$$V = \int_0^{2\pi} \int_{\pi/6}^{\pi/4} \int_0^4 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \boxed{\frac{64\pi}{3}(\sqrt{3} - \sqrt{2})}.$$

Q3(a). $\mathbf{r}'(t) = \langle 3, 4, 5 \rangle$, so $ds = 5\sqrt{2} \, dt$, while

$$x + y + z = 12t.$$

Hence

$$\oint_C (x + y + z) \, ds = 60\sqrt{2} \int_0^1 t \, dt = \boxed{30\sqrt{2}}.$$

Q3(b). $P = 2x - y^2$, $Q = x^2 + 3y$, hence

$$Q_x - P_y = 2x + 2y.$$

The triangular region is $0 \leq x \leq 2$, $0 \leq y \leq x$. Thus

$$\oint_C P \, dx + Q \, dy = \int_0^2 \int_0^x (2x + 2y) \, dy \, dx = \boxed{8}.$$

Q4. $z = g(x, y) = 4 - x - y$, so the upward area vector is

$$\langle -g_x, -g_y, 1 \rangle dA = \langle 1, 1, 1 \rangle dA.$$

Hence

$$\mathbf{F} \cdot \mathbf{n} \, dS = [y - x + (4 - x - y)^2] dA.$$

The domain is symmetric under interchange of x and y , so the integral of $y - x$ is zero. Therefore

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \int_0^4 \int_0^{4-x} (4 - x - y)^2 \, dy \, dx = \boxed{\frac{64}{3}}.$$

Mock Final 5

Q1(a). Let

$$T = 100e^{-(x^2+2y^2+z^2)}.$$

Then

$$\nabla T = 100e^{-(x^2+2y^2+z^2)} \langle -2x, -4y, -2z \rangle.$$

At $P = (1, 1, 0)$,

$$\nabla T = 100e^{-3} \langle -2, -4, 0 \rangle.$$

The direction from P to $Q = (3, 0, 2)$ is $\frac{1}{3} \langle 2, -1, 2 \rangle$, so

$$D_{\mathbf{u}}T = \boxed{0}.$$

Q1(b). Max/min of $f = x + y$ subject to $x^2 + 2y^2 = 6$:

$$\langle 1, 1 \rangle = \lambda \langle 2x, 4y \rangle.$$

Thus $x = 1/(2\lambda)$, $y = 1/(4\lambda)$, and the constraint gives $\lambda = \pm 1/4$. Therefore

$$\boxed{f_{\max} = 3 \text{ at } (2, 1)}, \quad \boxed{f_{\min} = -3 \text{ at } (-2, -1)}.$$

Q2(a). In polar coordinates,

$$I = \int_0^{2\pi} \int_1^2 \frac{r}{1+r^2} dr d\theta = 2\pi \left[\frac{1}{2} \ln(1+r^2) \right]_1^2 = \boxed{\pi \ln \frac{5}{2}}.$$

Q2(b). On the upper half-ball $\rho \leq 3$,

$$\iiint_E \rho dV = \int_0^{2\pi} \int_0^{\pi/2} \int_0^3 \rho^3 \sin \phi d\rho d\phi d\theta = \boxed{\frac{81\pi}{2}}.$$

Q3(a). For $\mathbf{r}(t) = \langle t, \cos t, \sin t \rangle$,

$$|\mathbf{r}'(t)| = \sqrt{2}, \quad yz = \cos t \sin t.$$

Thus

$$\int_C yz ds = \sqrt{2} \int_0^{\pi/2} \cos t \sin t dt = \boxed{\frac{\sqrt{2}}{2}}.$$

Q3(b). $P = x + y^2$, $Q = x^2 - y$, so

$$Q_x - P_y = 2x - 2y.$$

The enclosed region is $0 \leq x \leq 1$, $x^2 \leq y \leq x$. Therefore

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^1 \int_{x^2}^x (2x - 2y) dy dx = \boxed{\frac{1}{30}}.$$

Q4. On the lateral cylinder $x^2 + y^2 = 4$, the outward unit normal is

$$\mathbf{n} = \left\langle \frac{x}{2}, \frac{y}{2}, 0 \right\rangle.$$

For $\mathbf{v} = \langle x, y, 2z \rangle$,

$$\mathbf{v} \cdot \mathbf{n} = \frac{x^2 + y^2}{2} = 2.$$

Parametrize by $\mathbf{r}(\theta, z) = \langle 2 \cos \theta, 2 \sin \theta, z \rangle$, so $dS = 2 d\theta dz$. Hence

$$\iint_S \mathbf{v} \cdot \mathbf{n} dS = \int_0^3 \int_0^{2\pi} 4 d\theta dz = \boxed{24\pi}.$$

Since the flux is positive for the outward orientation, the side surface has net outward flow.